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Studierichting: Informatica
Oefeningenreeks 5G nr. 6.6

$f(x) = -2x^2 + 6x$

$g(x) = x^2$

$f(x) = g(x)$

$-2x^2 + 6x = x^2$

$\Leftrightarrow -3x^2 + 6x = 0$

$\Leftrightarrow -3x(x-2) = 0$

$\Leftrightarrow x = 0$ of $x = 2$

$\Rightarrow x \in [0, 2]$

We testen een punt tussen 0 en 2: $x = 1$

$f(1) = -2(1)^2 + 6(1) = 4$

$g(1) = 1^2 = 1$

$\Rightarrow f(x)$ is de bovenste functie

$$\Rightarrow V = \pi \int_0^2 \left(f(x)^2 - g(x)^2\right) dx$$
$$= \pi \int_0^2 \left((-2x^2 + 6x)^2 - (x^2)^2\right) dx$$

$(-2x^2 + 6x)^2 = 4x^4 - 24x^3 + 36x^2$

$(x^2)^2 = x^4$

$\Rightarrow (-2x^2 + 6x)^2 - (x^2)^2 = 4x^4 - 24x^3 + 36x^2 - x^4$

$= 3x^4 - 24x^3 + 36x^2$

$$\Rightarrow V = \pi \int_0^2 (3x^4 - 24x^3 + 36x^2) dx$$
$$= \pi \left[\frac{3x^5}{5} - \frac{24x^4}{4} + \frac{36x^3}{3}\right]_0^2$$
$$= \pi \left[\frac{3x^5}{5} - 6x^4 + 12x^3\right]_0^2$$

$$= \pi \left(\frac{3(2)^5}{5} - 6(2)^4 + 12(2)^3 \right) - \pi \left(\frac{3(0)^5}{5} - 6(0)^4 + 12(0)^3 \right)$$
$$= \pi \left(\frac{96}{5} - 96 + 96 \right) - \pi(0)$$
$$= \pi \cdot \frac{96}{5}$$
$$\Rightarrow V = \frac{96\pi}{5}$$

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References